



Service Tool Instruction

Service Tool Instruction
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2004

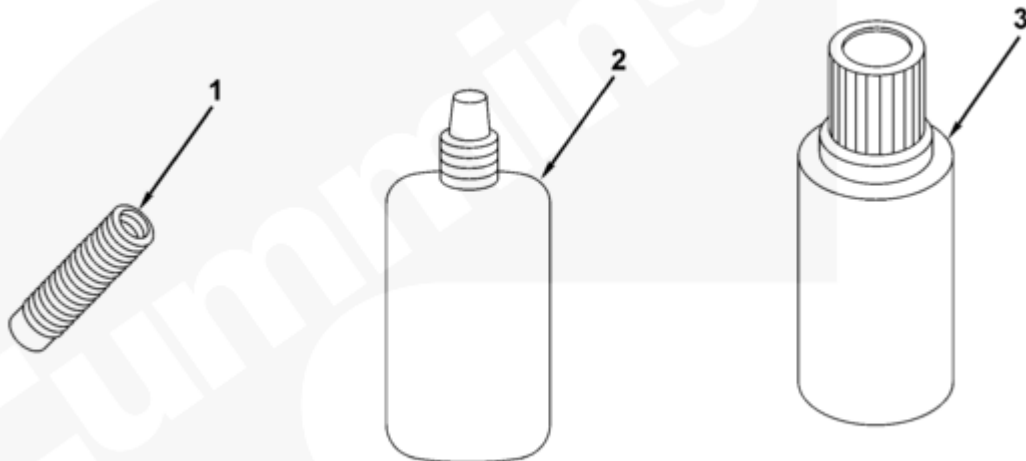
Description

Metric Thread Insert Kit

Purpose

This document provides information for repairing damaged metric threads using metric thread insert kit, Part Number 3822709. The kit is used with alignment fixture, Part Number ST 1232.

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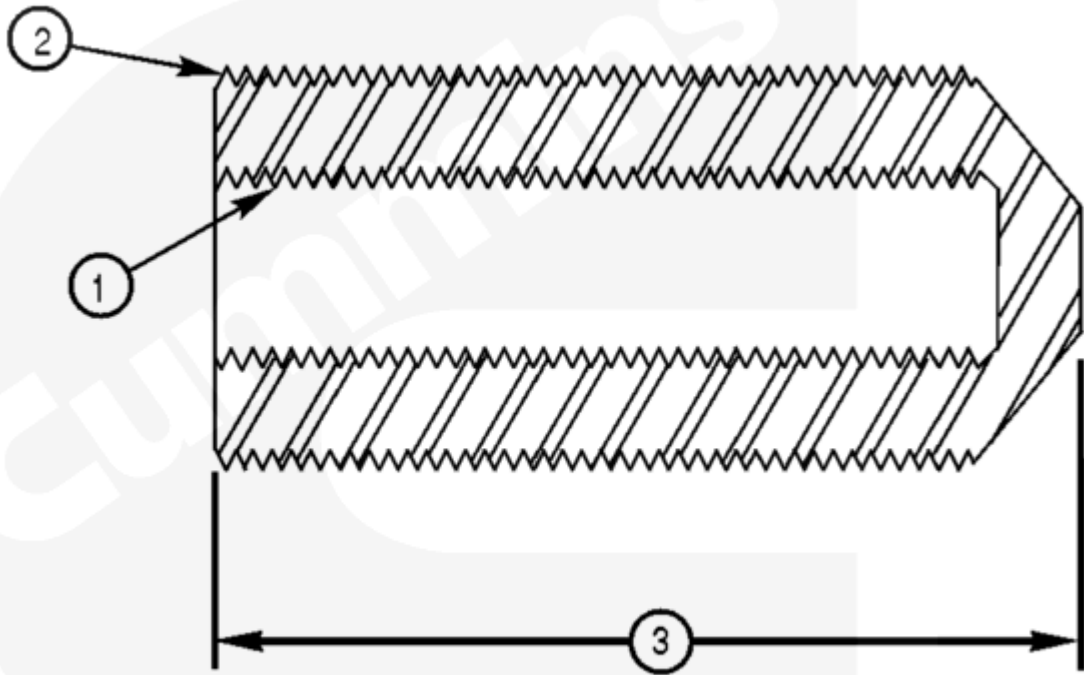
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Table 1, Metric Thread Insert Kit, Part Number 3822709

Item Number	Part Number	Description	Quantity
1	3822698	Insert, M6 x 1.0 (1 shown)	18
1	3822699	Insert, M8 x 1.25 (1 shown)	18
1	3822700	Insert, M10 x 1.5 (1 shown)	18
1	3822701	Insert, M12 x 1.25 (1 shown)	12
1	3822702	Insert, M12 x 1.5 (1 shown)	12

Table 1, Metric Thread Insert Kit, Part Number 3822709			
Item Number	Part Number	Description	Quantity
1	3822703	Insert, M12 x 1.75 (1 shown)	12
1	3822704	Insert, M14 x 2.0 (1 shown)	6
1	3822705	Insert, M16 x 1.5 (1 shown)	6
1	3822706	Insert, M16 x 2.0 (1 shown)	6
2	3824038	Thread-locking compound	1
3	3824715	Primer	1
Not shown	3163757	Storage case	1
Not shown	3377556	Kit content bulletin	1

Table 2, Items Used with Metric Thread Insert Kit, Part Number 3822709, Purchased Separately			
Item Number	Part Number	Description	Quantity
Not shown	ST 1232	Drill/ream fixture	1
Not shown	3164019	Chip vacuum	1
Not shown	See Table 3	Drill bushing	1



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Insert

**Table 3,
Thread
Insert
Dimensions**

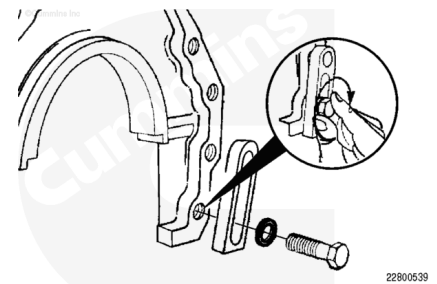
Part Number	Internal Thread Size (1)	External Thread Size (2)	Drill Size	Overall Length (3)	Drill Bushing
3822698	M6 x 1.0	3/8-16	5/16 (0.3125)	3/4-in	4918170
3822699	M8 x 1.25	1/2-13	27/64 (0.421)	15/16-in	4918172
3822700	M10 x 1.5	9/16-12	31/64 (0.484)	1-1/8-in	3376495
3822701	M12 x 1.25	5/8-18	37/64 (0.578)	1-3/8-in	ST-1234
3822702	M12 x 1.5	5/8-18	37/64 (0.578)	1-3/8-in	ST-1234
3822703	M12 x 1.75	5/8-18	37/64 (0.578)	1-3/8-in	ST-1234
3822704	M14 x 2.0	3/4-16	11/16 (0.687)	1-1/8-in	ST-1236

Table 3, Thread Insert Dimensions					
Part Number	Internal Thread Size (1)	External Thread Size (2)	Drill Size	Overall Length (3)	Drill Bushing
3822705	M16 x 1.5	7/8-14	13/16 (0.812)	1-1/4-in	4918173
3822706	M16 x 2.0	7/8-14	13/16 (0.812)	1-1/4-in	4918173

Install the correct spacers and washers onto the fixture alignment capscrews.

Install the capscrew and spacer assemblies through the mounting slot of the fixture.

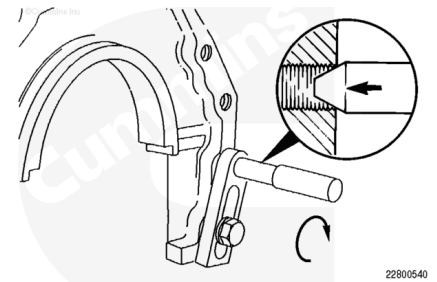
Finger-tighten the fixture mounting capscrew.



Use a tapered pin to align the guide bar and bushing over the damaged capscrew thread.

Push downward on the tapered pin and tighten the fixture mounting capscrew.

Torque Value: 67 n·m [49 ft-lb]



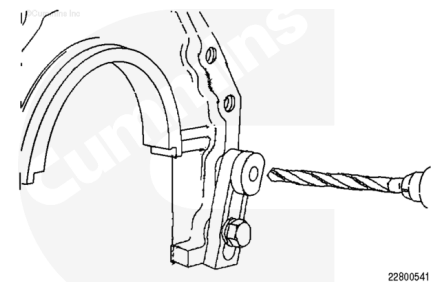
Using the information in Table 3, obtain the correct drill size.

Install the appropriate-size drill/ream bushing.

If necessary, mark the drill with tape for the required drill depth before starting the drilling operation.

Drill to the bottom of the original capscrew hole.

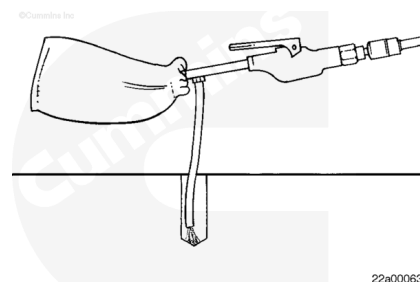
Remove the alignment fixture.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use the chip vacuum attached to compressed air to remove the chips from the newly drilled capscrew hole.

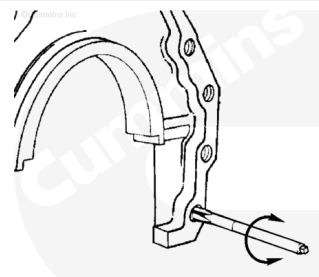


Note : Stop frequently to clean the bore using a chip vacuum.

Using the information contained in Table 3, External Thread Size, obtain correct tap size.

Tap the full depth of the hole using a suitable tapping fluid.

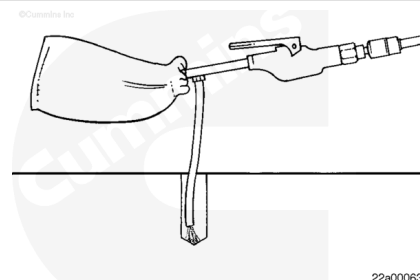
Remove the tap.



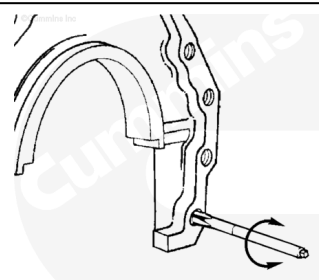
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use the chip vacuum attached to compressed air to remove the chips from the newly tapped capscrew hole.

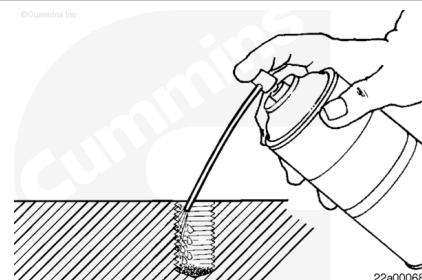


Thread the appropriate tap in and out of the newly cut threads several times to verify the thread condition, and clean debris from the threads.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

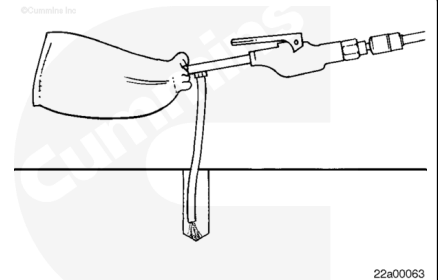
Note : Check the capscrew hole for evidence of porosity and cracks.

Clean the threads, and flush any debris from the tapped hole using a solvent-degreasing agent.

⚠ WARNING ⚠

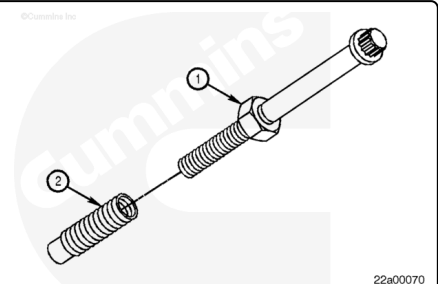
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use the chip vacuum attached to compressed air to remove the chips and solvent-degreasing agent from the newly tapped capscrew hole.



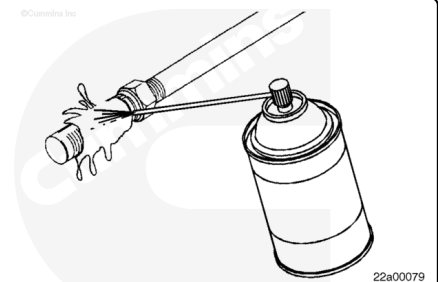
Place the appropriate jam nut (1) on a capscrew.

Install the appropriate thread salvage insert (2) onto the capscrew until the capscrew nears the bottom of the insert. Rotate the jam nut (1) until it contacts the thread salvage insert (2).



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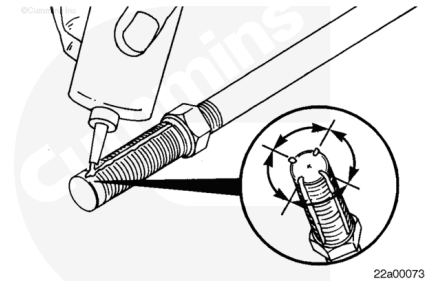
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Remove the preservative coating from the circumference of the thread salvage insert using a solvent-degreasing agent.

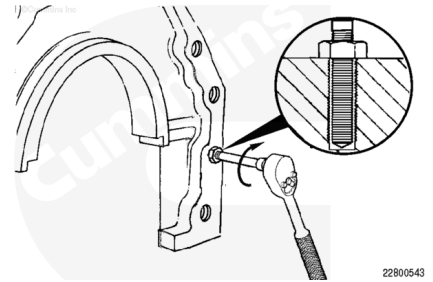
Apply a light coat of Primer to the thread insert and the threaded hole; allow 3 to 5 minutes to dry.

Apply four beads of thread-locking compound, Part Number 3824038, to the circumference of the thread salvage insert. Each bead **must** be 0.8 mm [0.03 in] wide down the entire length of the thread salvage insert, and the beads **must** be 90 degrees apart.



Install the insert until it is flush with the block or it bottoms in the hole.

Allow to dry for 3 hours.



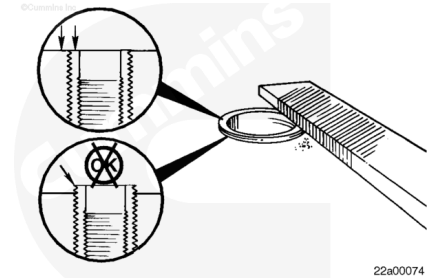
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove the capscrew and jam nut.

If necessary, file or machine the top of the thread salvage insert so that it is flush with the surface.

Make certain that the inner diameter of the thread salvage insert is free of burrs. Remove any file shavings or debris with the chip vacuum.

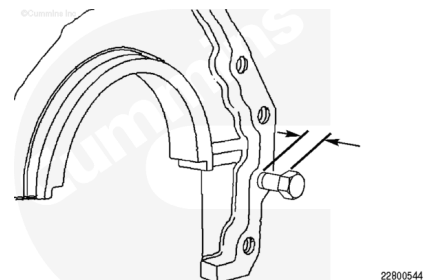


Note : It is important that sufficient thread engagement be maintained after this repair.

Install a replacement capscrew until it bottoms out in the repaired hole.

Measure the distance from under the capscrew head to the component's surface. Make certain that this dimension is a minimum of 2 mm [0.079 in] less than the thickness of the component and gasket to be attached.

If the distance is within tolerance, the installation is complete. If the distance is **not** within tolerance, the capscrew can be shortened slightly, or a shorter capscrew can be used. If a



shorter capscrew is used, make certain that a minimum thread engagement of 1-1/2 times the capscrew diameter is maintained.

Last Modified: 02-Feb-2004
